

Sample Input/Output:

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=====
NION ORCHESTRATION MAP
=====
Message: MSG-001
From: Sarah Chen (Product Manager)
Project: PRJ-ALPHA

=====
L1 PLAN
=====
[TASK-001] → L2:TRACKING_EXECUTION
  Purpose: Extract action items from customer request

[TASK-002] → L2:TRACKING_EXECUTION
  Purpose: Extract risks from scope change request

[TASK-003] → L2:TRACKING_EXECUTION
  Purpose: Extract decision needed

[TASK-004] → L3:knowledge_retrieval (Cross-Cutting)
  Purpose: Retrieve project context and timeline

[TASK-005] → L2:COMMUNICATION_COLLABORATION
  Purpose: Formulate gap-aware response
  Depends On: TASK-001, TASK-002, TASK-003, TASK-004

[TASK-006] → L3:evaluation (Cross-Cutting)
  Purpose: Evaluate response before sending
  Depends On: TASK-005

[TASK-007] → L2:COMMUNICATION_COLLABORATION
  Purpose: Send response to sender
  Depends On: TASK-006

=====
L2/L3 EXECUTION
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[TASK-001] L2:TRACKING_EXECUTION
  ↳ [TASK-001-A] L3:action_item_extraction
    Status: COMPLETED
    Output:
      • AI-001: "Evaluate real-time notifications"
        Owner: ? | Due: ? | Flags: [MISSING_OWNER, MISSING_DUE_DATE]
      • AI-002: "Evaluate dashboard export feature"
        Owner: ? | Due: ? | Flags: [MISSING_OWNER, MISSING_DUE_DATE]
```

```
[TASK-002] L2:TRACKING_EXECUTION
↳ [TASK-002-A] L3:risk_extraction
Status: COMPLETED
Output:
  • RISK-001: "Adding multiple features within the same timeline"
    Likelihood: HIGH | Impact: HIGH
  • RISK-002: "Scope creep from expanded requirements"
    Likelihood: MEDIUM | Impact: MEDIUM

[TASK-003] L2:TRACKING_EXECUTION
↳ [TASK-003-A] L3:decision_extraction
Status: COMPLETED
Output:
  • DEC-001: "Accept or reject the request described in the message"
    Decision Maker: ? | Status: PENDING

[TASK-004] L3:knowledge_retrieval (Cross-Cutting)
Status: COMPLETED
Output:
  • Project: PRJ-ALPHA
  • Current Release Date: Dec 15
  • Days Remaining: 20
  • Code Freeze: Dec 10
  • Current Progress: 70%
  • Team Capacity: 85% utilized
  • Engineering Manager: Alex Kim
  • Tech Lead: David Park

[TASK-005] L2:COMMUNICATION_COLLABORATION
↳ [TASK-005-A] L3:qna
Status: COMPLETED
Output:
  • Response: "Thanks for the update, Sarah! For real-time notifications and dashboard export feature:

    WHAT I KNOW:
    • Current timeline: Dec 15 (code freeze Dec 10)
    • Team capacity: 85% utilized
    • Progress: 70% complete

    WHAT I'VE LOGGED:
    • 2 action items for feature evaluation
    • Risk(s) flagged for timeline/scope impact
    • 1 pending decision

    WHAT I NEED:
    • Complexity estimates from Engineering (Alex Kim / David Park)
    • Go/no-go decision from leadership

    I cannot assess feasibility without Engineering input on whether this can fit in the remaining timeline at current capacity."
```

```
[TASK-006] L3:evaluation (Cross-Cutting)
Status: COMPLETED
Output:
  • Relevance: PASS
  • Accuracy: PASS
  • Tone: PASS
  • Gaps Acknowledged: PASS
  • Result: APPROVED

[TASK-007] L2:COMMUNICATION_COLLABORATION
↳ [TASK-007-A] L3:message_delivery
Status: COMPLETED
Output:
  • Channel: email
  • Recipient: Sarah Chen
  • CC: Alex Kim (Engineering Manager)
  • Delivery Status: SENT
```

=====

Test Case 1: Simple Status Question

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NION ORCHESTRATION MAP

=====

Message: MSG-101

From: John Doe (Engineering Manager)

Project: PRJ-BETA

=====

L1 PLAN

=====

[TASK-001] → L3:knowledge_retrieval (Cross-Cutting)

Purpose: Retrieve current status and context for requested item

[TASK-002] → L2:COMMUNICATION_COLLABORATION

Purpose: Formulate status response

Depends On: TASK-001

[TASK-003] → L3:evaluation (Cross-Cutting)

Purpose: Evaluate response before sending

Depends On: TASK-002

[TASK-004] → L2:COMMUNICATION_COLLABORATION

Purpose: Send response to sender

Depends On: TASK-003

=====

L2/L3 EXECUTION

=====

[TASK-001] L3:knowledge_retrieval (Cross-Cutting)

Status: COMPLETED

Output:

- Project: PRJ-BETA
- Current Release Date: Dec 15
- Days Remaining: 20
- Code Freeze: Dec 10
- Current Progress: 70%
- Team Capacity: 85% utilized
- Engineering Manager: Alex Kim
- Tech Lead: David Park

[TASK-002] L2:COMMUNICATION_COLLABORATION

↳ [TASK-002-A] L3:qna

Status: COMPLETED

Output:

- Response: "Hi John, here is the current status:
 - Progress: 70% complete
 - Team capacity: 85% utilized
 - On track for the Dec 15 release date (code freeze Dec 10)

Let me know if you need a more detailed breakdown."

[TASK-003] L3:evaluation (Cross-Cutting)

Status: COMPLETED

Output:

- Relevance: PASS
- Accuracy: PASS
- Tone: PASS
- Gaps Acknowledged: PASS
- Result: APPROVED

[TASK-004] L2:COMMUNICATION_COLLABORATION

↳ [TASK-004-A] L3:message_delivery

Status: COMPLETED

Output:

- Channel: slack
- Recipient: John Doe
- Delivery Status: SENT

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Test Case 2: Feasibility Question (New Features)

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NION ORCHESTRATION MAP

=====

Message: MSG-102

From: Sarah Chen (Product Manager)

Project: PRJ-ALPHA

=====

L1 PLAN

=====

[TASK-001] → L2:TRACKING_EXECUTION

Purpose: Extract action items from customer request

[TASK-002] → L2:TRACKING_EXECUTION

Purpose: Extract risks from scope change request

[TASK-003] → L2:TRACKING_EXECUTION

Purpose: Extract decision needed

[TASK-004] → L3:knowledge_retrieval (Cross-Cutting)

Purpose: Retrieve project context and timeline

[TASK-005] → L2:COMMUNICATION_COLLABORATION

Purpose: Formulate gap-aware response

Depends On: TASK-001, TASK-002, TASK-003, TASK-004

[TASK-006] → L3:evaluation (Cross-Cutting)

Purpose: Evaluate response before sending

Depends On: TASK-005

[TASK-007] → L2:COMMUNICATION_COLLABORATION

Purpose: Send response to sender

Depends On: TASK-006

=====

L2/L3 EXECUTION

=====

[TASK-001] L2:TRACKING_EXECUTION

↳ [TASK-001-A] L3:action_item_extraction

Status: COMPLETED

Output:

- AI-001: "Evaluate SSO integration"

Owner: ? | Due: ? | Flags: [MISSING_OWNER, MISSING_DUE_DATE]

[TASK-002] L2:TRACKING_EXECUTION

↳ [TASK-002-A] L3:risk_extraction

Status: COMPLETED

Output:

- RISK-001: "Adding SSO integration may impact existing release timeline"

Likelihood: MEDIUM | Impact: MEDIUM

```
[TASK-003] L2:TRACKING_EXECUTION
↳ [TASK-003-A] L3:decision_extraction
Status: COMPLETED
Output:
  • DEC-001: "Accept or reject the request described in the message"
    Decision Maker: ? | Status: PENDING

[TASK-004] L3:knowledge_retrieval (Cross-Cutting)
Status: COMPLETED
Output:
  • Project: PRJ-ALPHA
  • Current Release Date: Dec 15
  • Days Remaining: 20
  • Code Freeze: Dec 10
  • Current Progress: 70%
  • Team Capacity: 85% utilized
  • Engineering Manager: Alex Kim
  • Tech Lead: David Park

[TASK-005] L2:COMMUNICATION_COLLABORATION
↳ [TASK-005-A] L3:qna
Status: COMPLETED
Output:
  • Response: "Thanks for the update, Sarah! For SSO integration:

    WHAT I KNOW:
    • Current timeline: Dec 15 (code freeze Dec 10)
    • Team capacity: 85% utilized
    • Progress: 70% complete

    WHAT I'VE LOGGED:
    • 1 action item for feature evaluation
    • Risk(s) flagged for timeline/scope impact
    • 1 pending decision

    WHAT I NEED:
    • Complexity estimates from Engineering (Alex Kim / David Park)
    • Go/no-go decision from leadership

    I cannot assess feasibility without Engineering input on whether this can fit in the remaining timeline at current capacity."
```

```
[TASK-006] L3:evaluation (Cross-Cutting)
Status: COMPLETED
Output:
  • Relevance: PASS
  • Accuracy: PASS
  • Tone: PASS
  • Gaps Acknowledged: PASS
  • Result: APPROVED

[TASK-007] L2:COMMUNICATION_COLLABORATION
↳ [TASK-007-A] L3:message_delivery
Status: COMPLETED
Output:
  • Channel: email
  • Recipient: Sarah Chen
  • CC: Alex Kim (Engineering Manager)
  • Delivery Status: SENT
```

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Test Case 3: Decision/Recommendation Request

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NION ORCHESTRATION MAP
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Message: MSG-103
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From: Mike Johnson (VP Engineering)
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Project: PRJ-GAMMA
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```

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=====
L1 PLAN
=====
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[TASK-001] → L2:TRACKING_EXECUTION
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```
  Purpose: Extract decision needed and identify decision criteria
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```
[TASK-002] → L3:knowledge_retrieval (Cross-Cutting)
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```
  Purpose: Retrieve project context relevant to the decision
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```
[TASK-003] → L2:COMMUNICATION_COLLABORATION
```

```
  Purpose: Formulate recommendation/response
```

```
  Depends On: TASK-001, TASK-002
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```
[TASK-004] → L3:evaluation (Cross-Cutting)
```

```
  Purpose: Evaluate response before sending
```

```
  Depends On: TASK-003
```

```
[TASK-005] → L2:COMMUNICATION_COLLABORATION
```

```
  Purpose: Send response to sender
```

```
  Depends On: TASK-004
=====
```

```
=====
L2/L3 EXECUTION
=====
```

```
[TASK-001] L2:TRACKING_EXECUTION
```

```
  ↳ [TASK-001-A] L3:decision_extraction
```

```
    Status: COMPLETED
```

```
    Output:
```

- DEC-001: "Determine priority between competing initiatives"

```
      Decision Maker: ? | Status: PENDING
```

[TASK-002] L3:knowledge_retrieval (Cross-Cutting)

Status: COMPLETED

Output:

- Project: PRJ-GAMMA
- Current Release Date: Dec 15
- Days Remaining: 20
- Code Freeze: Dec 10
- Current Progress: 70%
- Team Capacity: 85% utilized
- Engineering Manager: Alex Kim
- Tech Lead: David Park

[TASK-003] L2:COMMUNICATION_COLLABORATION

↳ [TASK-003-A] L3:qna

Status: COMPLETED

Output:

- Response: "Hi Mike, based on current context:
 - Team capacity: 85% utilized, 70% through current scope
 - This decision has been logged as DEC-001 (PENDING)
- WHAT I NEED:
 - Risk/impact input from the relevant leads before recommending a priority order."

[TASK-004] L3:evaluation (Cross-Cutting)

Status: COMPLETED

Output:

- Relevance: PASS
- Accuracy: PASS
- Tone: PASS
- Gaps Acknowledged: PASS
- Result: APPROVED

[TASK-005] L2:COMMUNICATION_COLLABORATION

↳ [TASK-005-A] L3:message_delivery

Status: COMPLETED

Output:

- Channel: email
- Recipient: Mike Johnson
- Delivery Status: SENT

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Test Case 4: Meeting Transcript

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=====
NION ORCHESTRATION MAP
=====
Message: MSG-104
From: System (Meeting Bot)
Project: PRJ-ALPHA

=====
L1 PLAN
=====
[TASK-001] → L2:COMMUNICATION_COLLABORATION
  Purpose: Capture meeting transcript and generate minutes

[TASK-002] → L2:TRACKING_EXECUTION
  Purpose: Extract action items from meeting discussion

[TASK-003] → L2:TRACKING_EXECUTION
  Purpose: Extract issues/blockers raised in meeting

[TASK-004] → L2:COMMUNICATION_COLLABORATION
  Purpose: Generate meeting summary report
  Depends On: TASK-001, TASK-002, TASK-003

=====
L2/L3 EXECUTION
=====

[TASK-001] L2:COMMUNICATION_COLLABORATION
  ↳ [TASK-001-A] L3:meeting_attendance
    Status: COMPLETED
    Output:
      • Meeting captured: 4 speaker contributions logged
      • Dev: I'm blocked on API integration, staging is down.
      • QA: Found 3 critical bugs in payment flow.
      • Designer: Mobile mockups ready by Thursday.
      • Tech Lead: We might need to refactor the auth module.
      • Minutes generated and stored

[TASK-002] L2:TRACKING_EXECUTION
  ↳ [TASK-002-A] L3:action_item_extraction
    Status: COMPLETED
    Output:
      • AI-001: "Mobile mockups ready by Thursday."
        Owner: Designer | Due: ? | Flags: [MISSING_DUE_DATE]
      • AI-002: "We might need to refactor the auth module."
        Owner: Tech Lead | Due: ? | Flags: [MISSING_DUE_DATE]
```

[TASK-003] L2:TRACKING_EXECUTION

↳ [TASK-003-A] L3:issue_extraction

Status: COMPLETED

Output:

- ISSUE-001: "I'm blocked on API integration, staging is down." (raised by Dev)
Severity: MEDIUM | Status: OPEN
- ISSUE-002: "Found 3 critical bugs in payment flow." (raised by QA)
Severity: MEDIUM | Status: OPEN

[TASK-004] L2:COMMUNICATION_COLLABORATION

↳ [TASK-004-A] L3:report_generation

Status: COMPLETED

Output:

- Report Type: Meeting Summary
- Contributions Captured: 4
- Action Items Logged: see TRACKING_EXECUTION output
- Issues Logged: see TRACKING_EXECUTION output
- Report Status: GENERATED

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Test Case 5: Urgent Escalation

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=====
NION ORCHESTRATION MAP
=====
Message: MSG-105
From: Lisa Wong (Customer Success Manager)
Project: PRJ-DELTA

=====
L1 PLAN
=====
[TASK-001] → L2:TRACKING_EXECUTION
  Purpose: Extract escalated issue and assess severity

[TASK-002] → L3:knowledge_retrieval (Cross-Cutting)
  Purpose: Retrieve history and context for the escalated item

[TASK-003] → L2:COMMUNICATION_COLLABORATION
  Purpose: Formulate urgent response addressing escalation
  Depends On: TASK-001, TASK-002

[TASK-004] → L3:evaluation (Cross-Cutting)
  Purpose: Evaluate response tone and accuracy before sending
  Depends On: TASK-003

[TASK-005] → L2:COMMUNICATION_COLLABORATION
  Purpose: Send priority response to sender
  Depends On: TASK-004

=====
L2/L3 EXECUTION
=====

[TASK-001] L2:TRACKING_EXECUTION
  ↳ [TASK-001-A] L3:issue_extraction
    Status: COMPLETED
    Output:
      • ISSUE-001: "The client is asking why feature X promised for Q3 is still not delivered...."
        Severity: HIGH | Status: OPEN
```

[TASK-002] L3:knowledge_retrieval (Cross-Cutting)

Status: COMPLETED

Output:

- Project: PRJ-DELTA
- Current Release Date: Dec 15
- Days Remaining: 20
- Code Freeze: Dec 10
- Current Progress: 70%
- Team Capacity: 85% utilized
- Engineering Manager: Alex Kim
- Tech Lead: David Park

[TASK-003] L2:COMMUNICATION_COLLABORATION

↳ [TASK-003-A] L3:qna

Status: COMPLETED

Output:

- Response: "Hi Lisa, I understand the urgency. Here is what I found:

- Issue logged as ISSUE-001 with HIGH severity
- Pulling historical context on this item now

WHAT I NEED:

- Confirmation from the original owner on root cause and revised timeline before I can provide a full explanation."

[TASK-004] L3:evaluation (Cross-Cutting)

Status: COMPLETED

Output:

- Relevance: PASS
- Accuracy: PASS
- Tone: PASS
- Gaps Acknowledged: PASS
- Result: APPROVED

[TASK-005] L2:COMMUNICATION_COLLABORATION

↳ [TASK-005-A] L3:message_delivery

Status: COMPLETED

Output:

- Channel: email
- Recipient: Lisa Wong
- Priority: HIGH
- Delivery Status: SENT

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Test Case 6: Ambiguous Request

```
=====
NION ORCHESTRATION MAP
=====
Message: MSG-106
From: Random User (Unknown)
Project: N/A
=====

L1 PLAN
=====
[TASK-001] → L3:knowledge_retrieval (Cross-Cutting)
  Purpose: Attempt best-effort context retrieval (limited info available)

[TASK-002] → L2:COMMUNICATION_COLLABORATION
  Purpose: Formulate gap-aware clarifying response
  Depends On: TASK-001

[TASK-003] → L3:evaluation (Cross-Cutting)
  Purpose: Evaluate response before sending
  Depends On: TASK-002

[TASK-004] → L2:COMMUNICATION_COLLABORATION
  Purpose: Send clarifying response to sender
  Depends On: TASK-003

=====
L2/L3 EXECUTION
=====

[TASK-001] L3:knowledge_retrieval (Cross-Cutting)
  Status: COMPLETED
  Output:
    • Project: UNKNOWN (not specified in message)
    • Context retrieval limited - no project identifier available

[TASK-002] L2:COMMUNICATION_COLLABORATION
  ↳ [TASK-002-A] L3:qna
    Status: COMPLETED
    Output:
      • Response: "Hi Random, thanks for flagging this. The message doesn't specify which project, task, or area you mean.

        WHAT I NEED:
        • Which project this relates to
        • What specifically should be sped up (a feature, a process, a delivery date)

        Once I have that, I can pull the right context and give you a concrete answer."

[TASK-003] L3:evaluation (Cross-Cutting)
  Status: COMPLETED
  Output:
    • Relevance: PASS
    • Accuracy: PASS
    • Tone: PASS
    • Gaps Acknowledged: PASS
    • Result: APPROVED

[TASK-004] L2:COMMUNICATION_COLLABORATION
  ↳ [TASK-004-A] L3:message_delivery
    Status: COMPLETED
    Output:
      • Channel: slack
      • Recipient: Random User
      • Delivery Status: SENT
=====
```